

10.1

```
value.serializer = class org.apache.kafka.common.serialization.StringSerializer;

2026-02-21T10:21:40.492-08:00 INFO 55144 --- [Spring-Producer-Consumer] [nio-8088-exec-2] o.a.k.c.t.i.KafkaMetricsCollector : initializing Kafka metrics collector
2026-02-21T10:21:40.496-08:00 INFO 55144 --- [Spring-Producer-Consumer] [nio-8088-exec-2] o.a.k.c.producer.KafkaProducer : [Producer clientId=Spring-Producer-Consum
2026-02-21T10:21:40.501-08:00 INFO 55144 --- [Spring-Producer-Consumer] [nio-8088-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka version: 3.7.1
2026-02-21T10:21:40.501-08:00 INFO 55144 --- [Spring-Producer-Consumer] [nio-8088-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka commitId: e2494e6ffb89f828
2026-02-21T10:21:40.501-08:00 INFO 55144 --- [Spring-Producer-Consumer] [nio-8088-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka startTimeMs: 1771698188581
2026-02-21T10:21:40.505-08:00 INFO 55144 --- [Spring-Producer-Consumer] [umer-producer-1] org.apache.kafka.clients.Metadata : [Producer clientId=Spring-Producer-Consum
2026-02-21T10:21:40.506-08:00 INFO 55144 --- [Spring-Producer-Consumer] [umer-producer-1] o.a.k.c.p.internals.TransactionManager : [Producer clientId=Spring-Producer-Consum
Consumer Thread: 41 | Received: testMessage8
```

Consumers / consumer_group_1					
State	Members	Assigned Topics	Assigned Partitions	Coordinator ID	Total lag
STABLE	6	1	3	3	0

- 1) consumer service prints on terminal
- 2) total lag on consumer group is 0, proving that the latest log has been committed

10.2

```
Received message: testMessage9 from group: consumer_group_1 with topic: chuwa-yyds
Received message: testMessage10 from group: consumer_group_1 with topic: chuwa-yyds
Received message: testMessage11 from group: consumer_group_1 with topic: chuwa-yyds
Received message: testMessage12 from group: consumer_group_1 with topic: chuwa-yyds
Consumer Thread: 41 | Received: testMessage13
Received message: testMessage14 from group: consumer_group_1 with topic: chuwa-yyds
Received message: testMessage15 from group: consumer_group_1 with topic: chuwa-yyds
Consumer Thread: 41 | Received: testMessage16
Received message: testMessage17 from group: consumer_group_1 with topic: chuwa-yyds
```

Partition ID	Replicas	First Offset	Next Offset	Message Count	
0	1, 2, 3	0	5	5	⋮
1	2, 3, 1	0	5	5	⋮
2	3, 1, 2	0	7	7	⋮

- 1) thread 41 has not received
- 2) only three consumers are active although there are six consumers

10.3

Group ID	Num Of Members	Num Of Topics	Consumer Lag	Coordinator	State
consumer_group_1	9	1	N/A	3	STABLE
consumer_group_2	2	1	N/A	1	STABLE

10.4

+	6	2	2/21/2026, 10:30:17	testKey17	testMessage17
+	5	0	2/21/2026, 10:45:27	testKey18	testMessage18

Partition ID	Replicas	First Offset	Next Offset	Message Count	
0	1, 2, 3	0	6	6	...
1	2, 3, 1	0	5	5	...
2	3, 1, 2	0	7	7	...

- 1) test17, partition 2, offset 6; test 18, partition 0, offset 5
- 2) in consumers, partition 2 next offset 7; partition 0 next offset 6

10.5 Delivery guarantees

a. At-Least-Once. Default. Stop server the offset won't increase. If server restarts, message is delivered again.

b. At-Most-Once. Fast but risk. Consumer commits offset immediately when it receives messages, even before code runs. Code change:

```
props.put(ConsumerConfig.ENABLE_AUTO_COMMIT_CONFIG, "true");
props.put(ConsumerConfig.AUTO_COMMIT_INTERVAL_MS_CONFIG, "100"); // Commit
almost instantly
```

c. Exactly-Once. This ensures that even if a message is retried, the end result happens only once. In spring boot, we need to make it transactional.

@Bean

```
public ConcurrentKafkaListenerContainerFactory<String, String>
kafkaListenerContainerFactory() {

    ConcurrentKafkaListenerContainerFactory<String, String> factory = new
    ConcurrentKafkaListenerContainerFactory<>();

    factory.setConsumerFactory(consumerFactory());

    // This makes the consumer part of a transaction
```

```
factory.getContainerProperties().setAckMode(ContainerProperties.AckMode.MANUAL_IM  
MEDIATE);  
  
    return factory;  
}
```

10.6

Add custom partitioner to filter word “priority”. It should go to partition 0.

	6	0	2/21/2026, 11:28:38	testKey20	priority message 20
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